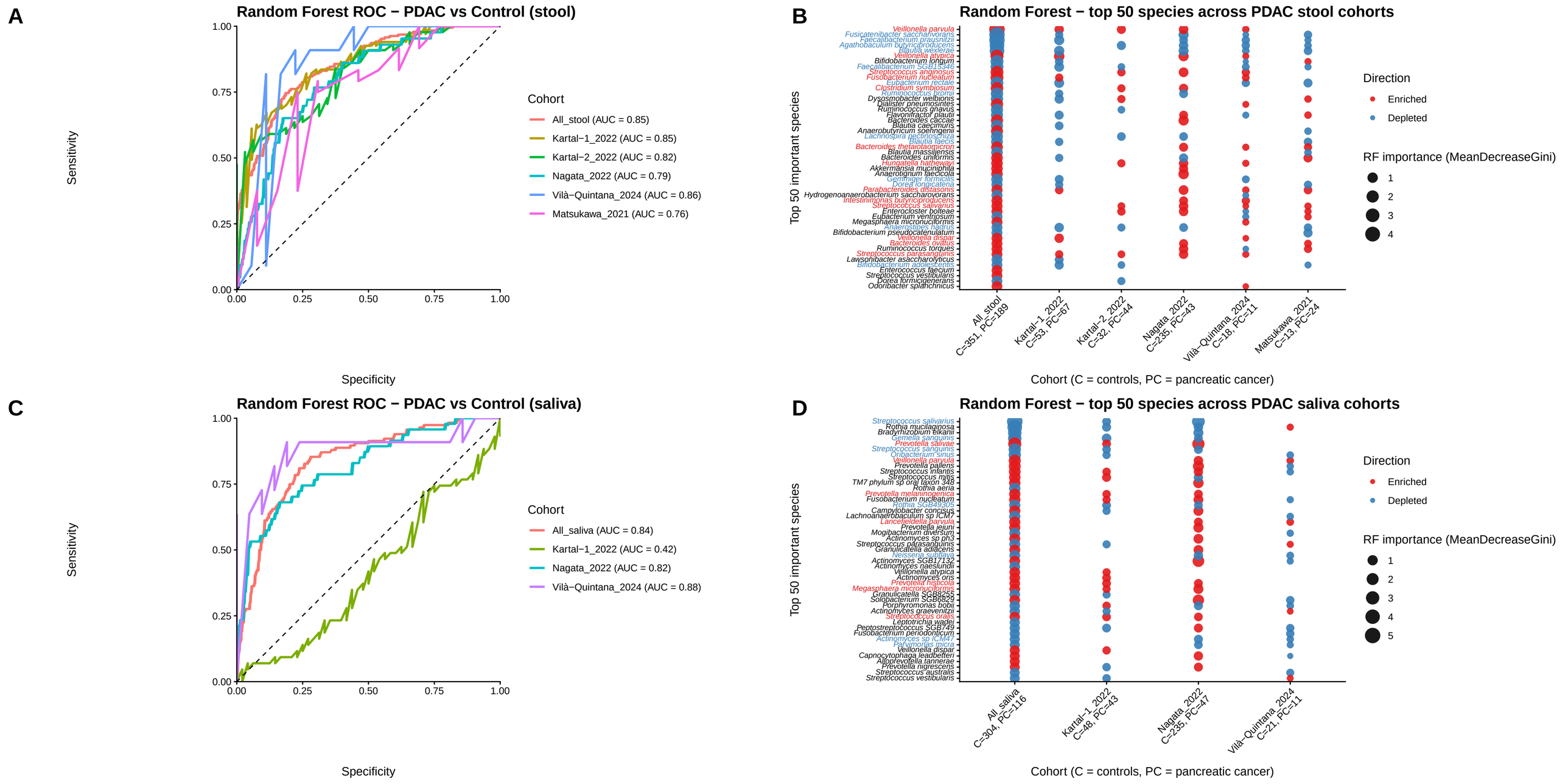
A
Random Forest ROC − PDAC vs Control (stool)
Sensitivity
Specificity
1.00
0.75
0.50
0.25
0.00
0.00
0.25
0.50
0.75
1.00
Cohort
All_stool (AUC = 0.85)
Kartal−1_2022 (AUC = 0.85)
Kartal−2_2022 (AUC = 0.82)
Nagata_2022 (AUC = 0.79)
Vilà−Quintana_2024 (AUC = 0.86)
Matsukawa_2021 (AUC = 0.76)
B
Random Forest − top 50 species across PDAC stool cohorts
Top 50 important species
Veillonella parvula
Fusicatenibacter saccharivorans
Faecalibacterium prausnitzii
Agathobaculum butyriciproducens
Blautia wexlerae
Veillonella atypica
Bifidobacterium longum
Faecalibacterium SGB15346
Streptococcus anginosus
Fusobacterium nucleatum
Eubacterium rectale
Clostridium symbiosum
Ruminococcus bromii
Dysosmobacter welbionis
Dialister pneumosintes
Ruminococcus gnavus
Flavonifractor plautii
Bacteroides caccae
Blautia caecimuris
Anaerobutyricum soehngenii
Lachnospira pectinoschiza
Blautia faecis
Bacteroides thetaiotaomicron
Blautia massiliensis
Bacteroides uniformis
Hungatella hathewayi
Akkermansia muciniphila
Anaerotignum faecicola
Gemmiger formicilis
Dorea longicatena
Parabacteroides distasonis
Hydrogenoanaerobacterium saccharovorans
Intestinimonas butyriciproducens
Streptococcus salivarius
Enterocloster bolteae
Eubacterium ventriosum
Megasphaera micronuciformis
Anaerostipes hadrus
Bifidobacterium pseudocatenulatum
Veillonella dispar
Bacteroides ovatus
Ruminococcus torques
Streptococcus parasanguinis
Lawsonibacter asaccharolyticus
Bifidobacterium adolescentis
Enterococcus faecium
Streptococcus vestibularis
Dorea formicigenerans
Odoribacter splanchnicus
All_stool C=351, PC=189
Kartal−1_2022 C=53, PC=67
Kartal−2_2022 C=32, PC=44
Nagata_2022 C=235, PC=43
Vilà−Quintana_2024 C=18, PC=11
Matsukawa_2021 C=13, PC=24
Cohort (C = controls, PC = pancreatic cancer)
Direction
Enriched
Depleted
RF importance (MeanDecreaseGini)
1
2
3
4
C
Random Forest ROC − PDAC vs Control (saliva)
Sensitivity
Specificity
1.00
0.75
0.50
0.25
0.00
0.00
0.25
0.50
0.75
1.00
Cohort
All_saliva (AUC = 0.84)
Kartal−1_2022 (AUC = 0.42)
Nagata_2022 (AUC = 0.82)
Vilà−Quintana_2024 (AUC = 0.88)
D
Random Forest − top 50 species across PDAC saliva cohorts
Top 50 important species
Streptococcus salivarius
Rothia mucilaginosa
Bradyrhizobium ikamii
Gemella sanguinis
Prevotella salivae
Streptococcus sanguinis
Oribacterium sinus
Veillonella parvula
Prevotella pallens
Streptococcus infantis
Streptococcus mitis
TM7 phylum sp oral taxon 348
Rothia aeria
Prevotella melaninogenica
Fusobacterium nucleatum
Rothia SGB49305
Campylobacter concisus
Lachnoanaerobaculum sp ICM7
Lancefieldella parvula
Prevotella jejuni
Mogibacterium diversum
Actinomyces sp ph3
Streptococcus parasanguinis
Granulicatella adiacens
Neisseria subflava
Actinomyces SGB17132
Actinomyces naeslundii
Veillonella atypica
Actinomyces oris
Prevotella histicola
Megasphaera micronuciformis
Granulicatella SGB8255
Solobacterium SGB6829
Porphyromonas bobii
Actinomyces graevenitzii
Streptococcus oralis
Leptotrichia wadei
Peptostreptococcus SGB749
Fusobacterium periodonticum
Actinomyces sp ICM47
Parvimonas micra
Veillonella dispar
Capnocytophaga leadbetteri
Alloprevotella tannerae
Prevotella nigrescens
Streptococcus australis
Streptococcus vestibularis
All_saliva C=304, PC=116
Kartal−1_2022 C=48, PC=43
Nagata_2022 C=235, PC=47
Vilà−Quintana_2024 C=21, PC=11
Cohort (C = controls, PC = pancreatic cancer)
Direction
Enriched
Depleted
RF importance (MeanDecreaseGini)
1
2
3
4
5